

ACTIVITY 3 - FUNCTIONS AND LOGICAL OPERATORS

Branch-name: js-activity-3



Activity 3 - FUNCTIONS

Task 1:

Instruction

- Start by defining four **functions**, one for each mathematical operation (addition, subtraction, multiplication, and division)

```
function add() {}
```

```
function subtract() {}
```

```
function multiple() {}
```

```
function divide() {}
```

- Add the correct parameters
- Each function should return the correct logic



Activity 3 - FUNCTIONS

Task 2:

Instruction

```
/*
```

```
    1. Write a function called greet that takes a name as an argument and returns a personalized greeting message.
```

```
*/
```

```
/*
```

```
    2. Write a function called calculateArea that takes the length and width of a rectangle as arguments and returns its area. The formula to calculate the area is: length * width.
```

```
*/
```

```
/*
```

```
    3. Write a function called celsiusToFahrenheit that takes a temperature value in Celsius and converts it to Fahrenheit. The formula to convert Celsius to Fahrenheit is: Fahrenheit = (Celsius * 9/5) + 32.
```

```
*/
```



Activity 3 - Increment & Decrement

Task 3:

Instruction

/*

1. Write a function called `incrementCounter` that takes a number as an argument, increases it by 1 using the `++` operator, and returns the new value.

*/

/*

2. Write a function called `decrementCounter` that takes a number as an argument, decreases it by 1 using the `--` operator, and returns the new value.

*/



Activity 3 - Comparison & Logical Operators

Task 4:

Instruction

```
/*
```

1. Write a function called `isAdult` that takes an age as an argument and returns true if the age is 18 or older, otherwise false.

```
*/
```

```
/*
```

2. Write a function called `canVote` that takes two arguments – age and `isCitizen` (boolean) – and returns true only if age is 18 or older AND `isCitizen` is true.

```
*/
```

```
/*
```

3. Write a function called `isInRange` that takes three numbers – num, min, max – and returns true if num is greater than or equal to min AND less than or equal to max.

```
*/
```



Activity 3 - Assignment + Comparison + Logical

Task 5:

Instruction

```
/*
```

1. Write a function called `applyDiscount` that takes two arguments – `price` and `discount` – and uses the `-=` assignment operator to subtract the discount from the price. Return the new price.

```
*/
```

```
/*
```

2. Write a function called `qualifiesForFreeShipping` that takes a `total` as an argument and returns `true` if the total is greater than or equal to 100, otherwise `false`.

```
*/
```



Activity 3 - Assignment + Comparison + Logical

Task 5:

Instruction

```
/*
```

```
3. Write a function called processOrder that takes three arguments - price,  
discount, and isMember (boolean). Apply the discount using -=. If the  
customer is a member OR the new price is at least 100, the order qualifies  
for free shipping. Return a string in the format:  
  
"Final price: <price>. Free shipping: <true/false>".
```

```
*/
```

